The Hong Kong Polytechnic University

Subject Description Form

Subject Code	AAE5112
Subject Title	Airport Operations and Management
Credit Value	3
Level	5
Pre-requisite/ Co-requisite/ Exclusion	Nil
Objectives	This subject will provide students with 1. a comprehensive understanding of the operational aspects of airports and how they are managed; 2. knowledge of the regulatory frameworks and safety standards that govern airport operations; 3. insight into the challenges of airport management including capacity, security, customer service, and technological advancements; 4. skills to analyse, design, and implement strategies for efficient and effective airport operations.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: a. understand and explain the key components of airport operations and the principles of airport management; b. analyse the impact of global and local regulatory frameworks on airport operations; c. evaluate the challenges in airport security, capacity, and customer service management; d. apply optimization methods to solve airport operational problems.
Subject Synopsis/ Indicative Syllabus	**Introduction to Airport Operations and Management**: Overview of airport infrastructure, stakeholders, and the role of airports in the aviation industry. **Airport Design and Capacity Management:** Factors influencing airport design, capacity planning, and strategies to optimize space and resources. **Airport Security Management**: Security regulations, risk assessment, and the implementation of security measures. **Technology in Airport Operations:** The role of information systems, automation, and emerging technologies in enhancing operational efficiency. **Operations Research in Airport Operations:** Optimization Methods that are typically used for solving operations management problems in airports, including runway scheduling, gate assignment, and aircraft taxiing routing.

Airport Business and Revenue Models: Understanding commercial activities, aeronautical vs. non-aeronautical revenues, and financial management.
Environmental Considerations: Approaches to managing environmental impacts, sustainability practices, and regulatory compliance.

Teaching/Learning Methodology

Teaching is conducted through class lectures and case studies. Lectures are aimed at delivering the core knowledge and concepts of airport operations and management and policy regulations relevant to airport operations. Case studies are to illustrate real-world applications and for in-depth discussions and problem-solving exercises.

Teaching/Learning Methodology	Intended subject learning outcomes to be covered			
	a	b	c	d
Lecture	√	√	√	√
Case study	√	√	√	√

Assessment Methods in Alignment with Intended Learning Outcomes

Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)			
		a	b	c	d
1. Assignment	30%			√	√
2. Group project	20%	√	√	√	√
3. Final examination	50%	√	√	√	√
Total	100 %				

Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes:

Overall Assessment:

$$0.5 \times \text{Continuous Assessment} + 0.5 \times \text{Final Examination}$$

The continuous assessment (50%) is aimed at enhancing the students' comprehension and assimilation of various topics of the syllabus via assignments and projects. The final examination (50%) will also be considered to assess the students' learning outcome.

Student Study Effort Expected

Class contact:	
▪ Lecture	39 Hrs.
Other student study effort:	
▪ Self-learning/preparation	36 Hrs.

	▪ Literature study/case study/reading	36 Hrs.
	Total student study effort	111 Hrs.
Reading List and References	1. Ashford, Norman J., H. P. Martin Stanton, Clifton A. Moore, Pierre Coutu, and John R. Beasley. Airport Operations. 3rd ed. McGraw-Hill Education, 2013. 2. Wells, Alexander T., and Seth B. Young. Airport Planning and Management, Sixth Edition, McGraw-Hill Education, 2011. 3. Kazda, Antonín, and Robert E. Caves, eds. Airport design and operation. Emerald Group Publishing Limited, 2010. 4. De Neufville, Richard. Airport systems planning, design, and management. In Air Transport Management, pp. 79-96. Routledge, 2020. 5. Chen, S., Wu, L., Ng, K. K., Liu, W., & Wang, K. How airports enhance the environmental sustainability of operations: A critical review from the perspective of Operations Research. Transportation Research Part E: Logistics and Transportation Review, 183, 103440, 2024.	

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